



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 16 July 2026 12:37 AM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This business report provides a comprehensive analysis of an employee dataset containing 21 records across 8 key variables. The primary objective is to evaluate demographic distributions, compensation structures, performance metrics, and general data health to support strategic HR decision-making. The dataset scores a high overall Health Score of 92, indicating strong structural integrity with minor quality anomalies. A notable finding is the near-perfect positive correlation (0.99) between employee age, experience, and salary. This suggests a rigid, tenure-based compensation structure. While this ensures predictable career progression, it introduces strategic risks regarding junior talent retention. This report details data quality findings, statistical relationships, and actionable recommendations to optimize both data management and workforce strategy. ---

DATASET OVERVIEW

The dataset consists of 21 rows and 8 columns, detailing employee profiles across several departments, including IT, HR, Finance, and Sales. The structural composition is defined as follows: * **Dataset Shape:** 21 rows, 8 columns * **Key Attributes:** * *Demographics:* Age,

Gender, City (including Bhubaneswar, Cuttack, Rourkela, Puri). * *Professional Metrics:* EmployeeID, Department, Experience (Years), Salary, Performance (Score). * **Preview Insights:** Employees range in age and experience (e.g., from a 24-year-old IT professional with 2 years of experience to a 35-year-old Finance professional with 10 years of experience). ---

DATA QUALITY ANALYSIS

An evaluation of the dataset's quality revealed a Health Score of 92. The following specific anomalies require attention before further analysis or modeling: * **Missing Values:** Exactly 1 missing value is present in the `Salary` column. * **Duplicate Rows:** There is 1 duplicate row within the dataset, which artificially inflates count-based metrics. * **Outliers:** Exactly 1 potential outlier has been flagged in the `Experience` column. * **Impact Assessment:** While these issues are minor, leaving them unaddressed will skew basic descriptive statistics (such as average salary and mean experience) and may distort predictive HR models. ---

STATISTICAL ANALYSIS

The dataset exhibits exceptionally strong linear correlations among its numerical variables: * **Age and Experience ($r = 0.99$):** A near-perfect correlation indicating that employee entry points into the organization or career track are highly standardized by age. * **Age/Experience and Salary ($r = 0.99$):** Salary is almost entirely determined by the age and tenure of the employee. There is a rigid salary progression model tied strictly to seniority. * **Performance and Salary/Experience ($r = 0.94$ & 0.92):** There is a highly positive correlation between experience and performance scores. Senior employees consistently receive the highest performance evaluations. * **Age and Performance ($r = 0.94$):** Older employees systematically achieve higher performance ratings. ---

BUSINESS INSIGHTS

* **Strictly Seniority-Based Culture:** The 0.99 correlation between experience and salary confirms that compensation is governed by tenure. There is little to no variation in salary that can be attributed to factors other than years on the job. * **Tenure-Biased Appraisals:** Performance scores are heavily correlated with experience (0.92). This suggests that either employees become significantly more productive as they age, or the performance appraisal process carries a strong subjective bias in favor of senior staff. * **Predictable Operational Footprint:** Operational presence is concentrated in key urban centers in Odisha (Bhubaneswar, Cuttack, Rourkela, Puri), reflecting a

localized regional talent pool. ---

RISK ASSESSMENT

* **Flight Risk of High-Performing Junior Talent:** Because compensation and performance scores are locked behind tenure, exceptional junior employees (e.g., high achievers with low experience) may feel undervalued. This can lead to high attrition among younger staff who seek merit-based compensation elsewhere. * **Data Multicollinearity:** For future data science projects (such as predicting employee turnover), the 0.99 correlation between `Age`, `Experience`, and `Salary` presents a severe multicollinearity risk. Using these variables together will destabilize regression-based machine learning models. * **Reporting Inaccuracies:** The presence of 1 duplicate row and 1 missing salary record compromise the accuracy of total payroll and headcount reporting. ---

CONCLUSION

The analyzed dataset represents a highly structured and standardized workforce. While the data health is strong (92), resolving the duplicate row, missing salary value, and experience outlier is necessary for clean reporting. Financially and operationally, the organization benefits from a predictable, tenure-aligned compensation structure. However, HR leadership must introduce merit-based flexibility to ensure that younger, high-performing talent is retained and motivated to grow within the company.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
Version	2.0.0
Generated On	16 July 2026 12:37 AM

DATASET SUMMARY

Metric	Value
Rows	21

Columns	8
Health Score	92%
Duplicate Rows	1

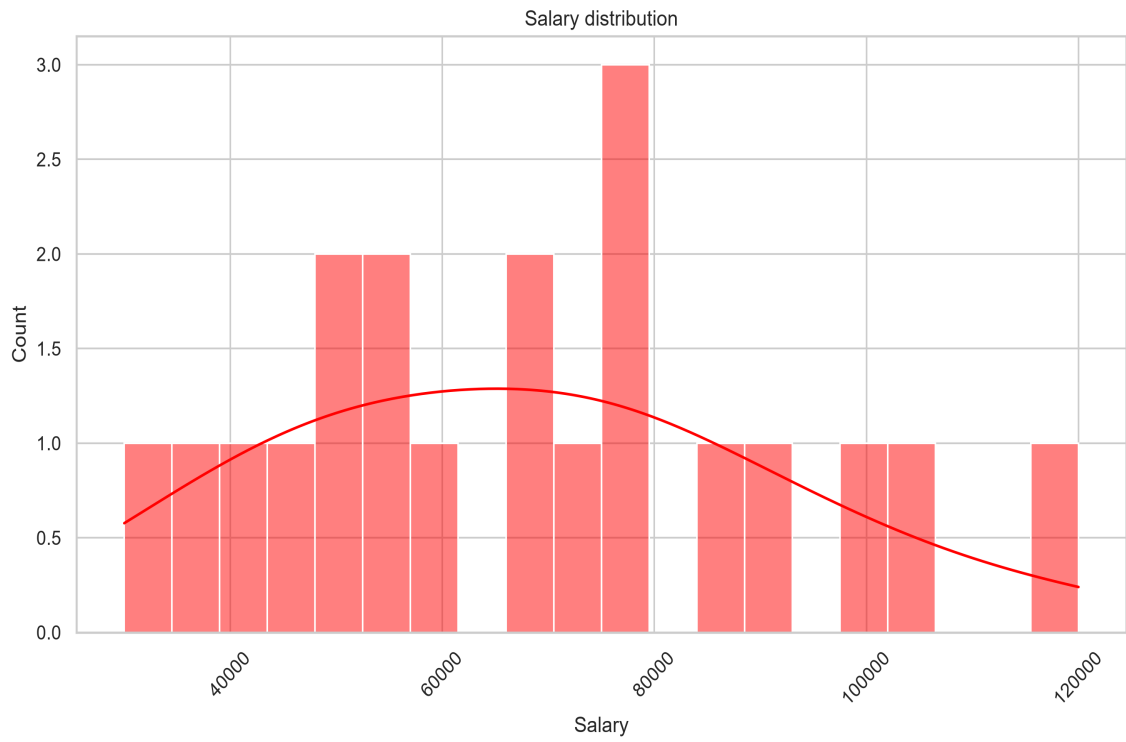
DATASET HEALTH SCORE

Overall Health Score : 92%

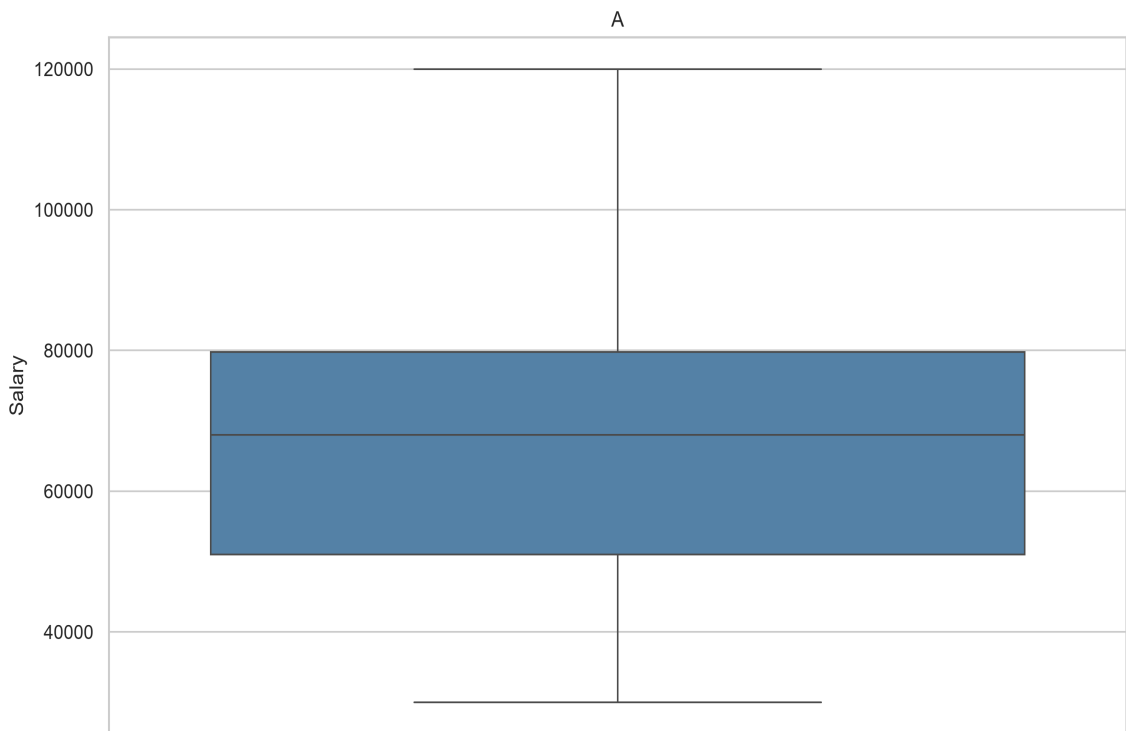
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

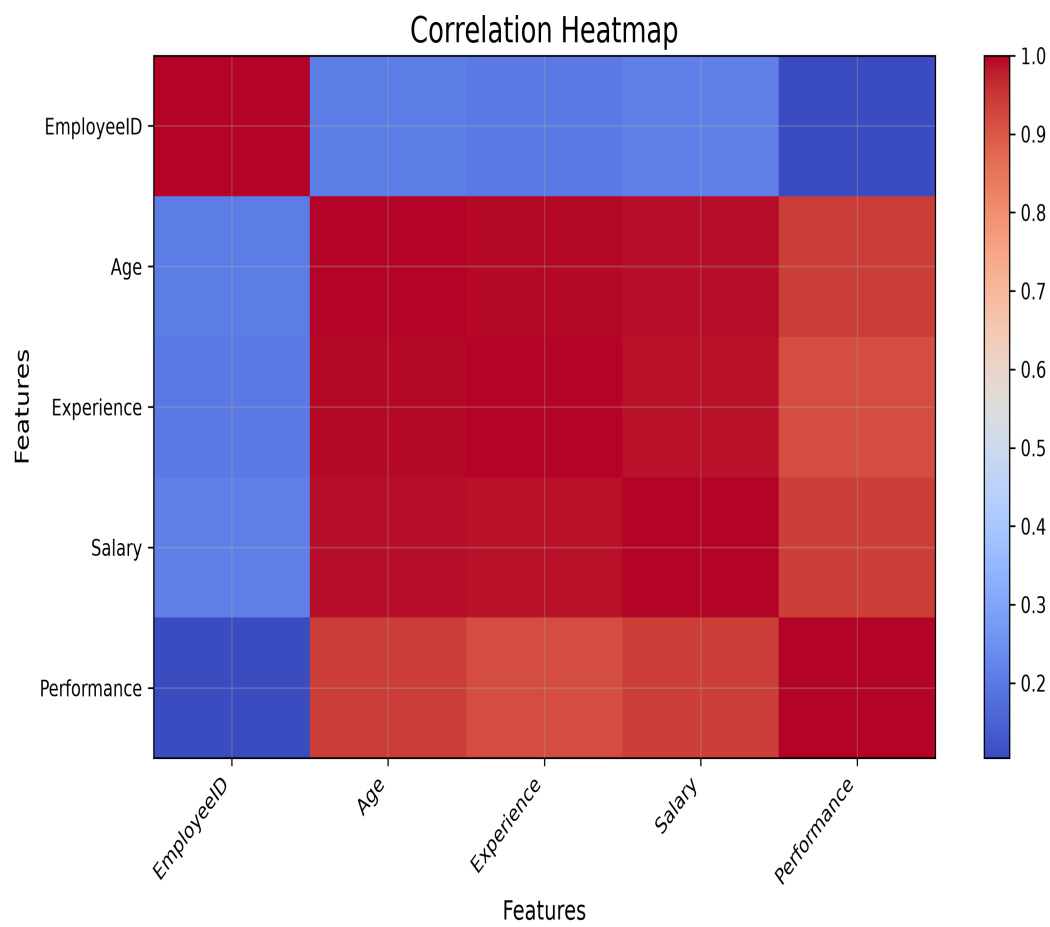
HISTOGRAM



BOX PLOT

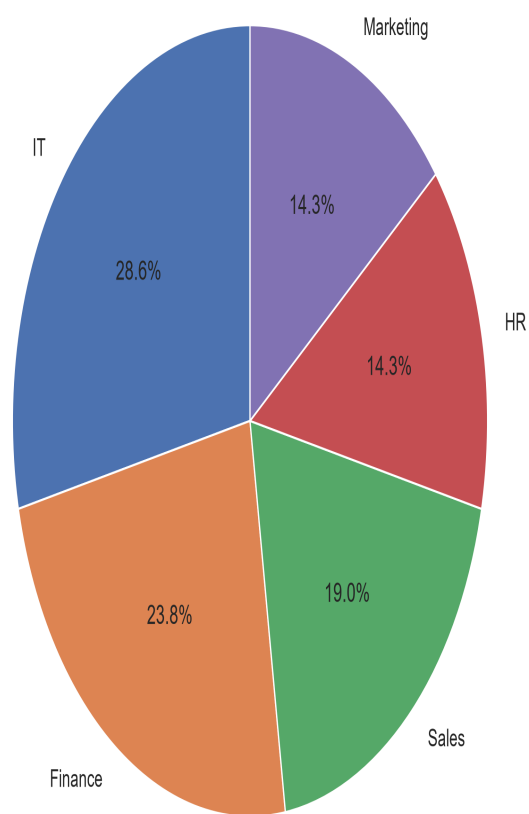


CORRELATION HEATMAP

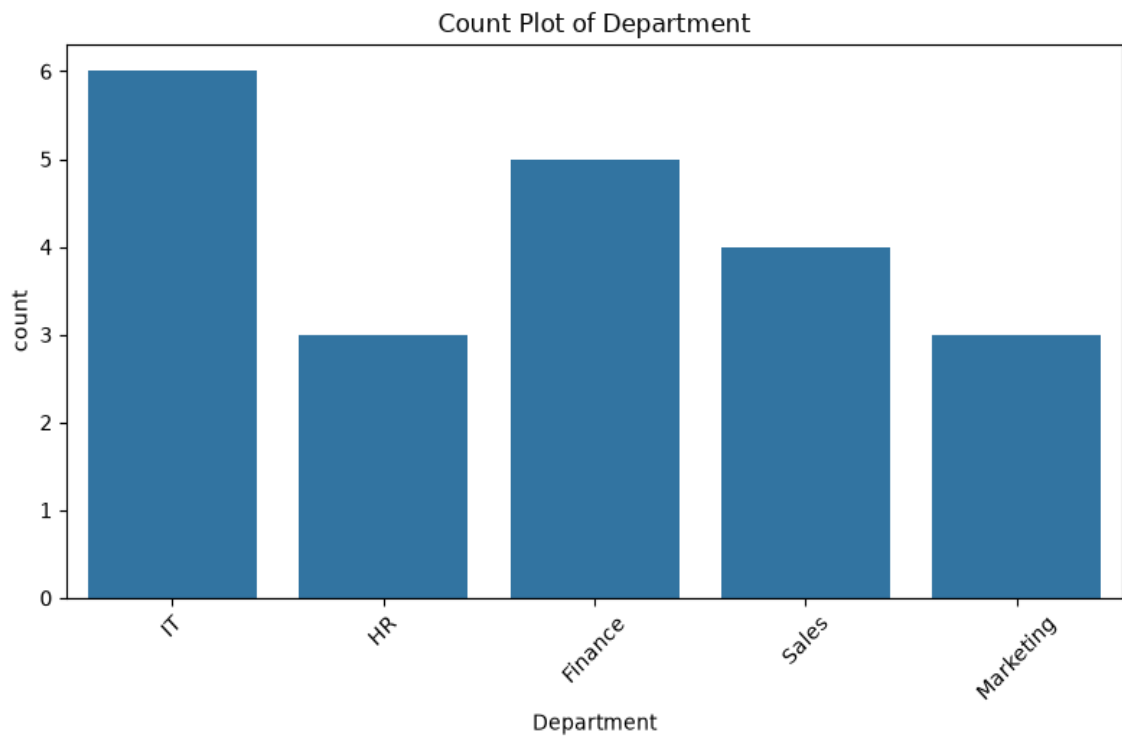


PIE CHART

Department Distribution



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

- ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ■■ 1 potential outliers detected.
- ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.